ALLIANCE UNIVERSITY
Private University established in Karnataka State by Act No.34 of year 2010
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NAAC GRADE A+
ACCREDITED UNIVERSITY

Alliance School of Advanced Computing Bengaluru

Department of CSE-IT

Capstone Project-I

Diary

Project/Batch ID	ASAC-CP-080		
Project Guide Name	Dr. Smitha Rajagopal		
Title of the Project	MedFedAI: Federated Intelligence for Privacy-Preserving Medical Image Classification		
Domain			
Team Leader Name	Pudi Pavani		
Details of the student	**Name**	**Registration Number**	**Branch**
	Pudi Pavani	2023BCSE07AED340	BTech-CSE-AIML
	Kurra Pothula Bhanu Gowtham	2023BCSE07AED344	BTech-CSE-AIML
	Sirivela Yashwanth Kumar	2023BCSE07AED347	BTech-CSE-AIML

Sign. of the Guide **Sign. of the Project Coordinator-Dept.** **Sign. of the Chief Project Coordinator-ASAC**

Project Details

Title of the Project:

MedFedAI: Federated Intelligence for Privacy-Preserving Medical Image Classification.

Abstract:

Artificial Intelligence (AI) is increasingly used for medical image analysis and classification. However, conventional centralized learning requires medical data from different healthcare institutions to be collected on a central server, creating privacy and security concerns. Moreover, medical data is often Non-Independent and Identically Distributed (Non-IID) due to differences in patient populations, imaging devices, data sizes, and class distributions. This project proposes a privacy-preserving Federated Learning framework for medical image classification under Non-Independent and Identically Distributed (Non-IID) data. Multiple simulated healthcare clients will perform local preprocessing and model training without sharing their raw medical images. The locally trained model parameters will be sent to a central server and aggregated using Federated Averaging (FedAvg) to create a global model. Different levels of data distribution will be considered to study their effect on model performance. The system will be evaluated using accuracy, precision, recall and F1-score and compared with conventional and standard Federated Learning approaches. The goal of this project is to understand the impact of Non-Independent and Identically Distributed (Non-IID) data on Federated Learning and explore a suitable approach for achieving privacy-aware and reliable medical image classification.

Daily Attendance Record

Student Name: Pudi Pavani **Reg. No: 2023BCSE07AED340**

Date	Day	Task Performed / Work Description	Present / Absent	Guide Signature

Daily Attendance Record

Student Name: Kurra Pothula Bhanu Gowtham Reg. No: 2023BCSE07AED344

Date	Day	Task Performed / Work Description	Present / Absent	Guide Signature

Daily Attendance Record

Student Name: Sirivela Yashwanth Kumar Reg. No: 2023BCSE07AED347

Date	Day	Task Performed / Work Description	Present / Absent	Guide Signature

Weekly Meeting Record

Week No: 01

Date: 13/08/26 to 21/08/26

Capstone Project Title: MedFedAI: Federated Intelligence for Privacy-Preserving Medical Image Classification

Group Members:

1. Name: Pudi Pavani Reg. No.2023BCSE07AED340 ☐ Present
2. Name: KurraPothula Bhanu Gowtham Reg. No.2023BCSE07AED344 ☐ Present
3. Name: Sirivela Yashwanth kumar Reg. No.2023BCSE07AED347 ☐ Present

Meeting Venue: LT212 ☐ On Time Documents

Submitted:

☐ Status Report

☐ On Time

Issues Group Working On:

Assessment of Progress:

☐ Excellent

☐ Good

☐ Satisfactory

☐ Fair

☐ Poor

Notes / Concerns / Comments: _____

Signature of the project Guide

Weekly Meeting Record

Week No: 02

Date: 22/08/26 to 28/08/26

Capstone Project Title: MedFedAI: Federated Intelligence for Privacy-Preserving Medical Image Classification

Group Members:

1. Name: Pudi Pavani Reg. No.2023BCSE07AED340 ☐ Present
2. Name: KurraPothula Bhanu Gowtham Reg. No.2023BCSE07AED344 ☐ Present
3. Name: Sirivela Yashwanth kumar Reg. No.2023BCSE07AED347 ☐ Present

Meeting Venue: LT212 ☐ On Time Documents

Documents Submitted:

☐ Status Report

☐ On Time

Issues Group Working On:

Assessment of Progress:

☐ Excellent

☐ Good

☐ Satisfactory

☐ Fair

☐ Poor

Notes / Concerns / Comments: _____

Signature of the project Guide

Weekly Meeting Record

Week No: 03

Date: 29/08/26 to 03/09/26

Capstone Project Title: MedFedAI: Federated Intelligence for Privacy-Preserving Medical Image Classification

Group Members:

1. Name: Pudi Pavani Reg. No.2023BCSE07AED340 ☐ Present
2. Name: KurraPothula Bhanu Gowtham Reg. No.2023BCSE07AED344 ☐ Present
3. Name: Sirivela Yashwanth Kumar Reg. No.2023BCSE07AED347 ☐ Present

Meeting Venue: LT212 ☐ On Time Documents

Documents Submitted:

☐ Status Report
☐ On Time

Issues Group Working On:

Assessment of Progress:

☐ Excellent
☐ Good
☐ Satisfactory
☐ Fair
☐ Poor

Notes / Concerns / Comments: _____

Signature of the project Guide

Weekly Meeting Record

Week No: 04

Date: 06/09/26 to 12/09/26

Capstone Project Title: MedFedAI: Federated Intelligence for Privacy-Preserving Medical Image Classification

Group Members:

1. Name: Pudi Pavani Reg. No.2023BCSE07AED340 ☐ Present
2. Name: KurraPothula Bhanu Gowtham Reg. No.2023BCSE07AED344 ☐ Present
3. Name: Sirivela Yashwanth Kumar Reg. No.2023BCSE07AED347 ☐ Present

Meeting Venue: LT212 ☐ On Time Documents

Submitted:

☐ Status Report

☐ On Time

Issues Group Working On:

Assessment of Progress:

☐ Excellent

☐ Good

☐ Satisfactory

☐ Fair

☐ Poor

Notes / Concerns / Comments: _____

Signature of the project Guide

Weekly Meeting Record

Week No: 05

Date: 14/09/26 to 19/09/26

Capstone Project Title: MedFedAI: Federated Intelligence for Privacy-Preserving Medical Image Classification

Group Members:

4. Name: Pudi Pavani Reg. No.2023BCSE07AED340 ☐ Present
5. Name: KurraPothula Bhanu Gowtham Reg. No.2023BCSE07AED344 ☐ Present
6. Name: Sirivela Yashwanth Kumar Reg. No.2023BCSE07AED347 ☐ Present

Meeting Venue: LT212 ☐ On Time Documents

Submitted:

☐ Status Report

☐ On Time

Issues Group Working On:

Assessment of Progress:

☐ Excellent

☐ Good

☐ Satisfactory

☐ Fair

☐ Poor

Notes / Concerns / Comments: _____

Signature of the project Guide

Weekly Meeting Record

Week No: 06

Date: 21/09/26 to 26/09/26

Capstone Project Title: MedFedAI: Federated Intelligence for Privacy-Preserving Medical Image Classification

Group Members:

7. Name: Pudi Pavani Reg. No.2023BCSE07AED340 ☐ Present
8. Name: KurraPothula Bhanu Gowtham Reg. No.2023BCSE07AED344 ☐ Present
9. Name: Sirivela Yashwanth Kumar Reg. No.2023BCSE07AED347 ☐ Present

Meeting Venue: LT212 ☐ On Time Documents

Submitted:

☐ Status Report

☐ On Time

Issues Group Working On:

Assessment of Progress:

☐ Excellent

☐ Good

☐ Satisfactory

☐ Fair

☐ Poor

Notes / Concerns / Comments: _____

Signature of the project Guide

Weekly Meeting Record

Week No: 07

Date: 28/09/26 to 03/10/26

Capstone Project Title: MedFedAI: Federated Intelligence for Privacy-Preserving Medical Image Classification

Group Members:

10. Name: Pudi Pavani Reg. No.2023BCSE07AED340 ☐ Present

11. Name: KurraPothula Bhanu Gowtham Reg. No.2023BCSE07AED344 ☐ Present

12. Name: Sirivela Yashwanth Kumar Reg. No.2023BCSE07AED347 ☐ Present

Meeting Venue: LT212 ☐ On Time Documents

Submitted:

☐ Status Report

☐ On Time

Issues Group Working On:

Assessment of Progress:

☐ Excellent

☐ Good

☐ Satisfactory

☐ Fair

☐ Poor

Notes / Concerns / Comments: _____

Signature of the project Guide

Weekly Meeting Record

Week No: 08

Date: 05/10/26 to 10/10/26

Capstone Project Title: MedFedAI: Federated Intelligence for Privacy-Preserving Medical Image Classification

Group Members:

13. Name: Pudi Pavani Reg. No.2023BCSE07AED340 ☐ Present

14. Name: KurraPothula Bhanu Gowtham Reg. No.2023BCSE07AED344 ☐ Present

15. Name: Sirivela Yashwanth Kumar Reg. No.2023BCSE07AED347 ☐ Present

Meeting Venue: LT212 ☐ On Time Documents

Submitted:

☐ Status Report

☐ On Time

Issues Group Working On:

Assessment of Progress:

☐ Excellent

☐ Good

☐ Satisfactory

☐ Fair

☐ Poor

Notes / Concerns / Comments: _____

Signature of the project Guide

Weekly Meeting Record

Week No: 09

Date: 12/10/26 to 17/10/26

Capstone Project Title: MedFedAI: Federated Intelligence for Privacy-Preserving Medical Image Classification

Group Members:

16. Name: Pudi Pavani Reg. No.2023BCSE07AED340 ☐ Present
17. Name: KurraPothula Bhanu Gowtham Reg. No.2023BCSE07AED344 ☐ Present
18. Name: Sirivela Yashwanth Kumar Reg. No.2023BCSE07AED347 ☐ Present

Meeting Venue: LT212 ☐ On Time Documents

Submitted:

☐ Status Report
☐ On Time

Issues Group Working On:

Assessment of Progress:

☐ Excellent
☐ Good
☐ Satisfactory
☐ Fair
☐ Poor

Notes / Concerns / Comments: _____

Signature of the project Guide

Review 1 Evaluation Form

Date of Review: _____

_____

Reviewer Comments: (*Provide specific feedback on the Problem Statement, Applications/Use Cases, Background Work, Feasibility Study, and overall direction of the project.)*

Recommendations for Improvement: (*List suggestions or required changes before the next phase/review.)*

Project Guide Signature	Examiner 1 Signature	Examiner 2 Signature
Name:	Name:	Name:
Date:	Date:	Date:

Review 2 Evaluation Form

Date of Review: _____

_____

Reviewer Comments: (*Provide specific feedback on the Abstract & Scope of the Project, Literature Survey, Project Requirements, and Presentation.)*

Recommendations for Improvement: (*List suggestions or required changes before the next phase/review.)*

Project Guide Signature	Examiner 1 Signature	Examiner 2 Signature
Name:	Name:	Name:
Date:	Date:	Date:

Semester End Exam Evaluation

Date of Review: _____

_____

Reviewer Comments: (*Provide specific feedback on the Project Report submitted, Problem Formulation, Review of Literature, Design Methodology, and Concept of the Work / Originality and Significance & Relevance – Application in the Real World.)*

_____

_____

_____

_____

_____

Project Submission Checklist

Item	Yes	No	Remarks
Project Report Submitted	☐	☐	
Paper Submitted in Conference or Journal	☐	☐	
Patent Filed or Granted	☐	☐	
Source Code / Software Submitted (if applicable)	☐	☐	
Prototype or Model Demonstrated (if applicable)	☐	☐	
Presentation Slides Submitted	☐	☐	
Project Diary Maintained	☐	☐	
Ethical Clearance Obtained (if applicable)	☐	☐	
Any Other Supporting Documents	☐	☐	

Examiner 1 Signature

Name:

Date:

Examiner 2 Signature

Name:

Date:

Student Consent Form

Academic Year: _____

Semester: _____

Degree Program: _____

Project Title: _____

Project Guide Name: _____

Guide's Designation: _____

Team Members

Sl. No.	Name of the Student	Register Number
1.		
2.		
3.		

Declaration by the Students

We hereby confirm that we have chosen the above topic for our Capstone Project – I as part of the curriculum for the _____semester of our _____degree program.

We understand and agree to the following:

1. The project work is academic in nature and must follow the guidelines provided by the department.
2. We will carry out the project work ethically and responsibly under the supervision of the assigned guide.
3. We will not engage in plagiarism or submit any copied or previously submitted work.
4. We will attend all review meetings and submit documentation as per the department schedule.
5. Any form of academic misconduct or non-compliance may lead to disqualification or disciplinary action.

Signatures of Students

Sl. No.	Name of the Student	Signature
1.		
2.		
3.		

Capstone Project – General Rules

1. All projects must be approved by the supervising faculty before starting.
2. All work must be original; plagiarism is strictly prohibited.
3. Students can work in teams.
4. Teams must have 2 to 4 members.
5. Interdepartmental (ASAC) teams are allowed and encouraged.
6. Roles and contributions of each team member must be clearly defined.
7. Teams must register with the Capstone Coordinator or supervisor by the deadline.
8. A team leader must be designated for communication and coordination.
9. Team members must contribute fairly and equally.
10. Conflicts must be reported promptly to the advisor.
11. Students must meet regularly with their faculty advisor (weekly or bi-weekly).
12. Meeting notes or progress reports must be maintained.
13. Follow official timelines for proposals, drafts, final reports, and presentations.
14. Late submissions may incur penalties unless extensions are granted.
15. Maintain detailed documentation including research sources, notes, code, and progress updates.
16. Projects must follow ethical guidelines (no falsified data, respect IP, consent/privacy for surveys).
17. Follow department formatting guidelines (APA, IEEE, etc.) for all submissions.
18. Submit all required documents: proposal, report, logbook, and presentation materials.
19. Deliver a final presentation and be prepared for questions.
20. Maintain professional conduct with advisors, teammates, and evaluators.
21. Demonstrate responsibility and a strong work ethic.
22. Use institutional resources responsibly and uphold academic integrity.